

Competition for order flow with fast and slow traders

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Abstract

The rise of computerized trading strategies in equity markets has spurred competition between trading venues. This paper shows that cross-venue strategies create highly interlinked markets: trades on one venue are followed by sizeable cancellations of limit orders on competing venues. These cancellations are explained in a simple model of competition between two limit order markets with fast and slow traders. Only the fast traders can access the liquidity of both venues simultaneously. Empirically, we confirm the predictions that the fraction of fast traders (i) reduces the equilibrium liquidity supply and (ii) reduces the magnitude of the cancellations following a trade.

JEL Codes: G10; G14; G15;

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1 Introduction

Two important trends have drastically changed the way equities are traded in recent years. First, technological innovations have led to high-frequency trading, a type of trading where computer algorithms analyze market data and trade at extremely high speed. Second, the market has become fragmented market as trading volume and liquidity are dispersed across multiple trading venues. These two changes in the structure of equity markets may strongly affect the trading strategies of investors with different trading speeds. In turn, the strategic interactions determine the correlations in liquidity across markets.¹

This paper finds that trades on one venue are immediately followed by cancellations of limit orders on competing venues of more than 60% of the trade size. These cancellations are explained in a model of competition between two limit order markets, with market makers and fast and slow traders. After the market makers observe a trade in one venue, they update their belief of the fundamental value and quickly cancel or revise limit orders on the same side of the order books of competing venues.

These cancellations following a trade are efficient and reflect the information content of the trade. This finding contrasts the view that high levels of cancellations are a symptom of market dysfunctionality. For example, regulators use the term “phantom-liquidity” to describe liquidity that disappears when needed by long-term investors (SEC, 2010). In a recent book, M. Lewis argues that high-frequency traders front-run slow traders: when a slow trader executes on one venue, before she can access a competing venue the high-frequency traders have already depleted all liquidity there (Lewis, 2014).

The first contribution is an analysis of the short-term correlations between the supply and demand of liquidity across trading venues. Specifically, I regress the change in the quoted depth at the bid or ask side of one venue on the buy and sell trading volumes of all

¹The high level of interaction between markets became apparent during the Flash Crash, that is, between E-Mini S&P 500 futures and individual stocks (SEC-CFTC, “Findings regarding the market events of May 6, 2010,” 2010).

individual venues. The analysis uses data of a sample of FTSE100 stocks containing the limit order books of the five relevant trading venues. These stocks are traded in a fairly fragmented environment, since the “incumbent” exchange—the London Stock Exchange (LSE)—executes 66% of the lit trading volume, leaving 34% to four “entrants.” The data are sampled at intervals of 100 milliseconds and the regressions include lagged values to measure the impact over time. To the best of my knowledge, this paper is the first to analyze the impact of high-frequency trading strategies on the supply and demand of liquidity across trading venues.

The empirical results strongly support the model’s main prediction that trades are followed by limit order cancellations on competing venues. That is, within 100 milliseconds, transactions on the three most active trading venues are followed by cancellations of limit orders on the *same side* of competing venues’ order books with a value of 29% to 67% of the transaction size. These effects strengthen with the time horizon and are as large as 38% to 84% of the transaction size after one second (depending on the trading venue). Further, trades are followed by submissions of limit orders on the *opposite side* of the order books with a value of 116% to 155% of the transaction size after one second.

The second contribution of the paper are tests of two novel predictions that relate the fraction of fast traders in the market to the magnitude of the cancellations and the consolidated quoted depth. In the model, only the fast traders can access the liquidity of all venues “simultaneously.” The slow traders effectively have access to only a single venue, because after an execution on one venue the market makers cancel their outstanding quotes on other venues before the slow trader can access these. In this setup, limit orders face a higher adverse selection risk when executing against a fast trader, even when fast and slow traders are equally informed on average. The reason is that, *conditional upon execution* of the limit order, a fast trader may have already traded on the competing venue and therefore the combined trades are larger and more informed on average. This risk does not arise with slow traders, since their lack of speed prevents them from accessing multiple

venues simultaneously.² This mechanism yields two key predictions. A higher fraction of fast traders (i) prompts the market makers to reduce their liquidity supply on all venues, and (ii) it reduces the magnitude of the cancellations following a trade.

A proxy for the fraction of fast traders is developed to test the key predictions of the model. In theory, fast traders submit two market orders to both venues simultaneously, such that the market makers are unable to revise or cancel their limit orders in between the two trades. Empirically, two trades are classified as simultaneous when the best bid and ask prices and quantities of the venue executing the current trade have not changed since the previous trade on a different venue. A first proxy of the fraction of fast traders is the percentage of pooled orders that occur simultaneously, which is 20% on average. This corresponds to 39% of all market orders, since a pooled order of a fast trader consists of at least two market orders sent to two venues. This measure depends on the quoted depth at time of trade, and a second proxy which corrects for this bias is 40% on average.

The empirical analysis supports the model's key predictions. First, a one standard deviation increase in the fraction of fast traders across firm-hours (9%) reduces the consolidated depth by 7%. Second, the average impact of an LSE buy on the consolidated ask-side depth is -1.43, implying that beyond the one-unit purchase, another 43% of the order size is canceled within one second. This impact changes to -1.32 after a one standard deviation increase in the fraction of fast traders. Thus, when the fraction of fast traders increases, the market makers offer less depth and cancel relatively fewer limit orders after a trade. The mechanics in the model are symmetrical for the bid and ask sides, and an LSE buy trade increases the consolidated depth on the bid sides by 1.16 units after one second. This impact reduces to 1.06 after a one standard deviation increase in the fraction of fast traders. The magnitudes are similar for LSE sell trades, and approximately 20%-50% larger

²This adverse selection risk follows from competition between trading venues, which is different from the channel that fast traders observe public information before slow traders (see Jovanovic and Menkveld (2011), Biais, Foucault, and Moinas (2011), Foucault, Hombert, and Rosu (2012), Hoffmann (2014) among others).

for trades on the entrants (depending on the specification).

The paper focusses on a single asset traded on multiple venues, but the tradeoffs in the model could be generalized to highly correlated assets. Further, the friction of speed heterogeneity applies to market settings other than the high-frequency trading environment. For example, in 1988 the Nasdaq introduced the Small Order Execution System (SOES) to provide automatic order execution, and a group of fast traders (so-called SOES bandits) started monitoring quotes of different dealers to pick off the slow ones (see the discussion in Foucault, Röell, and Sandås (2003)).

The theoretical model of this paper builds on the framework of a pure limit order market with adverse selection of Glosten (1994). It extends the specification of Sandås (2001) to a two-venue setting and contributes to Foucault and Menkveld (2008) by allowing for adverse selection.

The paper relates to the literature on competition between electronic limit order books. Glosten (1998) shows that two competing limit order markets can coexist when the tick size is discrete. The reason is that time priority is absent *across* venues, which raises competition between liquidity suppliers. Foucault and Menkveld (2008) further develop this point and call it the “queue-jumping” effect. Their model extends the framework of Parlour and Seppi (2003), who study competition between a pure limit order market and a hybrid market. Competition between exchanges can further arise through differences in the tick size (Biais, Bisière, and Spatt, 2010, Buti, Rindi, Wen, and Werner, 2011). Baruch, Karolyi, and Lemmon (2007) model competition between two batch auctions, where market makers are restricted to operate on one venue. This setting is suitable for cross-listing shares, but differs from the current setting where the market makers are active on all venues. Empirically, liquidity generally improves by competition between exchanges (O’Hara and Ye, 2011), but it may decrease by competition from opaque markets (Degryse, de Jong, and van Kervel, 2014).

2 The model

This section first describes the cancellation hypothesis. Then, it quantifies this cancellation effect by placing the problem into an adverse selection framework and derives testable implications. A simple numerical example concludes the section.

2.1 Cancellation hypothesis

In a fragmented trading environment, time priority is not enforced between trading venues, whereas price priority is enforced only when the trader has access to all venues.³ Price priority implies that limit orders with a better price are executed before those with a worse price, while time priority entails that limit orders placed first are executed first.

Due to the absence of time priority across venues, I hypothesize that limit order traders have an incentive to increase their liquidity provision across several venues. Then, after a trade on one venue, they will cancel limit orders on others, because the trade is informative about the fundamental value. By increasing the liquidity supply across venues, limit order traders increase their execution probability and expected profits. A tradeoff arises, however, since there is a risk that limit orders on several venues will be executed simultaneously. In this case, the liquidity supplier trades too much, which is costly in this model because of adverse selection risk. Therefore, liquidity suppliers have an incentive to minimize the risk of simultaneous executions by investing in high-frequency technology.

Who could pursue a strategy of increasing and canceling the liquidity provision with limit orders across venues? Arguably, financial intermediaries such as high-frequency market makers can strongly benefit from the increased execution probability. At the same time, using state-of-the-art technology, they monitor several venues in real time and may quickly cancel limit orders to reduce the risk of simultaneous executions. In contrast, this

³In the United States, price priority across markets is enforced by law (Regulation National Market System). Time priority however, crucial for this hypothesis, is not enforced.

strategy is not attractive to other traders. Those with an intrinsic motive to trade a large quantity would not cancel limit orders after a single execution. For others, the technology to monitor continuously is too costly.

2.2 Model setup

The goal is to study the equilibrium liquidity supply on two competing venues A and B in an adverse selection framework. The trading game proceeds as follows. Consider two types of investors: market makers and traders. The market makers are risk neutral and supply liquidity by placing limit orders on both venues. They are profit maximizing, competitive and have access to all venues. The traders demand liquidity by placing market orders. They may have private information about the fundamental value or liquidity motives and want to trade quickly. The market orders are informed on average and therefore impose adverse selection risk on the market makers. This setup is consistent with [Brogaard, Hendershott, and Riordan \(2014\)](#), who show that the market orders are informed on average, while limit orders of high-frequency traders get adversely selected.

The timeline of the trading game is presented in Figure 1. Trading occurs sequentially over periods indexed by t and each period consists of three stages. First, the market makers arrive and place limit orders on both venues. They do so until a competitive equilibrium is reached in which no market maker finds it optimal to place additional limit orders.⁴ Second, a trader arrives who places market orders with total size x , and, depending on the trader's type, executes limit orders on venue A, B, or both. Third, the market makers observe the executed trade and update their expectation of the fundamental value. Now, the game starts over and is repeated for every trade.

⁴See [Baruch and Glosten \(2013\)](#) for a related model with a mixed-strategy equilibrium liquidity supply and [Zhou Yueshen \(2014\)](#) for a model that focuses on how the equilibrium liquidity supply is obtained dynamically.

venues. The parameters α , β , and γ are exogenous, since we focus on a high-frequency trading environment.

Three reasons explain why some investors are not able to trade on both venues simultaneously. First, human traders may have access to both venues but trade too slowly, leaving a delay of several milliseconds when they split up a trade across two venues. In this case, high-frequency market makers have sufficient time to update their limit order schedules in between the trades. This high-frequency trading strategy is known as latency arbitrage, and segments the market according to trading speed. As a result, human traders effectively have access to one venue at a time.

In the United States, so-called inter-market sweep orders (ISOs) are designed to access liquidity on several venues, but they do not require a simultaneous arrival time of market orders (see (Chakravarty, Jain, Upson, and Wood, 2012) for a discussion). For this reason, the Royal Bank of Canada introduced an order routing technology that takes the routing latency of each venue into account to synchronize the arrival times of trades to all venues. A second source of millisecond delays stems from active searches for hidden liquidity. In this case, a trader sends a (marketable) limit order to one venue and must wait for the response to confirm whether hidden liquidity is executed before choosing the quantity to trade at another venue. This delay may give high-frequency traders sufficient time to update their limit orders in between the trades. Third, fast trading technology might be too costly to implement for some traders, since it requires fixed costs for technological infrastructure, software, programmers, and access to and the analysis of data feeds.

The trader is a buyer or seller with equal probability and has a reservation price $p_m > p_1$ at which she is not willing to buy.⁶ The average order size is the same for traders of all

⁶This small assumption prevents the trade from walking up the limit order book too much in the case of a thin order book, but does not affect the outcome of the model.

types, with mean ϕ and exponential density function

$$f(x) = \frac{1}{\phi} \exp\left(-\frac{x}{\phi}\right) \quad \text{if } x > 0 \text{ (market buy).} \quad (1)$$

The cumulative distribution function is $F(\cdot)$. A shortcoming of the model is that the order size is assumed to be exogenous. In a full equilibrium model, the order size would depend on the liquidity in the order book, the trader's expectation of the fundamental value, and her current inventory (see Biais, Martimort, and Rochet (2000) and Back and Baruch (2013)).

2.4 The market makers

Market makers place limit orders on venues A and B and know that trades are informative about the assets' next period fundamental value V_{t+1} . Market makers therefore face adverse selection risk, which will determine the equilibrium liquidity supply. The market makers monitor trades on both venues and update their expectation of the fundamental value using a linear price impact function

$$E(V_{t+1}|x_t) = E(V_t) + \lambda x_t. \quad (2)$$

The coefficient $\lambda > 0$, since purchases typically contain positive information with respect to the fundamental value (and similarly, sales contain negative information). Thus, larger orders cause more adverse selection costs and a greater price impact. In the remaining analysis the time subscript t is omitted.

The market makers place limit orders on a discrete pricing grid $\{\dots p_{-1}, p_0, p_1, p_2 \dots\}$. The model is symmetric, but for expositional reasons we focus on the ask side (prices $\{p_1, p_2 \dots\}$), where p_1 is the lowest ask price level that attracts a positive number of

shares. Denote the number of shares offered on venue $j \in \{A, B\}$ at each price level k by $\{Q_{j1}, Q_{j2}, \dots, Q_{jk}\}$.

Given the price impact function, we can calculate the expected profit of a one-share limit order placed at any location q in the queue of limit orders on each venue. The expected profit depends on the expected value of the asset conditional upon execution of the limit order $E(V|x \geq q)$ (the “upper-tail expectation” of (Glosten, 1994)). Note that while trades by fast and slow traders have the same size distribution and price impact parameter, market makers still face higher adverse selection costs when trading against fast traders. The reason is that, conditional upon the execution of a limit order, the fast traders have also traded on the competing venue and therefore the trade is larger and more informed on average.

2.5 Equilibrium

The model is in equilibrium when no market maker can profitably place an additional limit order at any price level (as in Sandås (2001) and Parlour and Seppi (2003)). Therefore, the expected profit of the *marginal limit order*, i.e., the single share offered at the end of the queue of limit orders, must equal zero for all price levels on both venues. This condition determines the equilibrium liquidity supply.

Proposition 1 *The solution for the equilibrium quantities Q_{A1} and Q_{B1} exists and is unique:*

$$Q_1^S \equiv \frac{p_1 - E(V) - \lambda\phi}{\lambda}, \quad (3)$$

$$Q_{A1} = Q_1^S - \frac{\gamma}{\gamma + \alpha(1 - \gamma) \exp(\frac{Q_{B1}}{\phi})} Q_{B1}, \quad (4)$$

$$Q_{B1} = Q_1^S - \frac{\gamma}{\gamma + (1 - \alpha)(1 - \gamma) \exp(\frac{Q_{A1}}{\phi})} Q_{A1}. \quad (5)$$

Proof. See the Appendix. ■

Equation (3) is the definition of Q_1^S , the optimal quantity offered in the single venue setting of Sandås (2001).⁷ By definition, p_1 is the best price level that attracts a positive number of shares, such that $Q_1^S > 0$. In (4), we observe that the equilibrium quantity on venue A equals Q_1^S minus a term that reflects the additional adverse selection cost of trading with a fast trader. Intuitively, this cost depends on the likelihood of the event times the expected loss in case of the event. The likelihood is represented by the fraction of the second term, which is the probability of trading with a fast trader given that the marginal limit order executes. The loss in case of the event is linear in Q_{B1} (the last term), which follows from the linear price impact function. Thus, for a market maker on venue A, an increase in the number of shares on venue B reduces the likelihood of trading with a fast trader but increases the cost in case of this event. The definition is analogous for venue B.

2.6 Testable implications

This section discusses three predictions of the model: a static prediction that holds at each point in time and two dynamic predictions that compare equilibrium liquidity before and after a trade.

The static prediction follows from the solution in equations (4) and (5).

Proposition 2 *If $\alpha = 0$ or $\beta = 0$ one venue attracts all liquidity and the single-exchange solution of Sandås (2001) obtains. Otherwise both venues attract a strictly positive number of shares at p_1 and are said to coexist. When both markets coexist,*

1. *An increasing fraction of fast traders (γ) strictly reduces the consolidated depth.*
2. *The consolidated depth is strictly larger than that of the single-exchange solution.*

⁷Sandås also adds a fixed limit order execution cost, which is easily added but does not yield new insights in a two venue setting.

The single-exchange equilibrium obtains when one venue attracts only fast investors (when $\alpha = 0$ or $\beta = 0$), in which case all liquidity diverts to the competing venue. The reason is that the marginal limit order faces higher adverse selection risk when executing against a fast trader: the fast trader also accessed the competing venue and therefore, conditional upon execution of the limit order, the trade is larger and more informed on average. Thus, a venue that attracts only fast investors faces strictly more adverse selection risk than its competitor, such that the competitor will attract all liquidity in equilibrium.

When both venues coexist, an increasing fraction of fast traders (γ) increases adverse selection risk, which in turn causes the market makers to reduce their liquidity supply. Foucault and Menkveld (2008) study a two-venue setting with smart order routers, similar to the fast traders in this model, and find that the relation between consolidated depth and the fraction of smart routers cannot be signed unambiguously. One mechanism in their model is the “queue-jumping” effect, which predicts that the fraction of smart routers *increases* the liquidity supply. In the current paper, for $\gamma = 0$, the introduction of a second venue doubles the quoted liquidity relative to the single exchange setting. Then, as γ increases the adverse selection cost raises and equilibrium depth declines. This paper contributes to Foucault and Menkveld (2008) by incorporating adverse selection risk, but does not have a fixed limit order submission cost.

The second part of the proposition states that when both venues coexist, the consolidated depth is higher than that of the single-exchange setting. In the special case in which all traders are fast ($\gamma = 1$), the market is virtually not fragmented and therefore the consolidated depth equals that of the single-exchange setting.⁸

The following two propositions relate to the impact of a trade on the equilibrium consolidated depth.

⁸If $\gamma = 1$, then Q_{A1} and Q_{B1} are no longer uniquely identified since any pair of (Q_{A1}, Q_{B1}) such that $Q_{A1} + Q_{B1} = Q_1^S$ is an equilibrium (see Equation (4)).

Proposition 3 Define the “Buy on Ask” impact (BoA) as the marginal impact of a buy market order on the consolidated quoted depth of the ask sides:

1. $BoA = \lambda \frac{\partial(Q_{A1} + Q_{B1})}{\partial E(V)}$.
2. In the single-exchange setting, $BoA = -1$.
3. When both markets coexist $BoA < -1$, and if $\gamma = 0$ then $BoA = -2$. The impact of a buy trade on the change in depth at the ask sides is larger than the trade size.
4. The BoA strictly increases in the fraction of fast traders (γ), i.e., becomes less negative, if the parameter values satisfy $Q_{A1} < \phi, Q_{B1} < \phi$.

The results for the ‘Sell on Bid’ impact variable (SoB) are identical to those for the BoA .

A one-unit trade increases the market makers’ expectation of the fundamental value by λ units (see (2)), who in turn update their equilibrium liquidity supply. Thus, the marginal impact of a buy trade on the equilibrium depth is defined by part (1).

In the single-exchange setting, a one-unit trade reduces liquidity by exactly one unit because the trade’s price impact equals the slope of the limit order book.⁹ When both markets coexist, the equilibrium depth is higher (Proposition 2), and a one-unit trade reduces the offered liquidity by more than one unit. Effectively, the private information of the trade is incorporated at the competing venue through cancellations of limit orders.

In the special case when no traders are fast ($\gamma = 0$), both venues are fully segmented and each obtain the liquidity of the single-exchange setting. In this case, a trade of x shares is followed by limit order cancellations of exactly x shares on the competing venue, such that $BoA = -2$. Then, increasing the level of γ will raise BoA , which moves towards -1 when $\alpha = 0$ or $\beta = 0$ and the single-exchange setting is obtained.¹⁰ The impact of γ on

⁹For a trade of size x (see equation (2)), the impact on Q_{A1} is $\frac{\Delta Q_{A1}}{x} = \frac{\Delta Q_{A1}}{\Delta V} \frac{\Delta V}{x} = -\frac{1}{\lambda} \lambda = -1$.

¹⁰The BoA is calculated as the *marginal* impact of a trade on quoted depth. If the trade x is large and depletes all quoted depth, the BoA is different. See details in the proof of Proposition 3 in the Appendix.

BoA is monotonically positive when the parameters are such that $Q_{A1} < \phi, Q_{B1} < \phi$.¹¹ Generally, the parameters are satisfied when the tick size is fairly small or the order size and price impact parameter are relatively large (which all directly affect Q_{A1} and Q_{B1}).

Proposition 4 *Define the “Buy on Bid” impact (BoB) as the marginal impact of a buy market order on the consolidated quoted depth on the bid sides:*

1. $BoB = \lambda \frac{\partial(Q_{A1}^{Bid} + Q_{B1}^{Bid})}{\partial E(V)}$.
2. *In the single-exchange setting, $BoB = 1$.*
3. *When both markets coexist $BoB > 1$, and if $\gamma = 0$ then $BoB = 2$. The impact of a buy trade on the change in depth at the bid sides is larger than the trade size.*
4. *The BoB strictly decreases in the fraction of fast traders (γ), if the parameter values satisfy $Q_{A1} < \phi, Q_{B1} < \phi$.*

This proposition is analogous to Proposition 3. The impact of a trade on the quoted depth is transmitted through the market makers’ expectation of the fundamental value, who update their quotes symmetrically across the bid and ask sides. By symmetry, the results for the impact of BoB are identical to those for the “Sell on Ask” impact (SoA).

Combining Propositions 3 and 4, a trade has an ambiguous effect on liquidity: A buy trade has a negative impact on ask-side liquidity but a positive impact on the bid-side liquidity. While the positive impact of a buy on the bid-side liquidity (part 1 of Proposition 4) is a basic prediction of the Glosten (1994) framework, parts 2 and 3 are specific contributions of this model. The symmetry between buys and sells and the bid and ask sides improve the robustness of the empirical tests of the mechanics in the model.

¹¹This parameter range is sufficient, but not necessary. See the Web appendix for details.

2.7 Numerical example

Figure 2 plots the equilibrium consolidated ask liquidity ($Q_{A1} + Q_{B1}$) against the fraction of fast traders (γ) before and after a one-unit buy trade. We consider the case where $\alpha = 0.5$ (dotted line) and where $\alpha = 0.8$ (dashed line). The solid line represents the liquidity of the single exchange setting, which is by definition unaffected by the fraction of fast traders. The parameters are substituted by realistic values, as described in the figure heading.

Figure 2 clearly shows that an increase in the fraction of fast traders reduces the equilibrium liquidity supply (Proposition 2). Further, the single-exchange setting yields the lowest level of liquidity (1.5 shares before the trade), which is reached when either $\alpha = 0$ or $\beta = 1 - \alpha = 0$. The liquidity is twice that of the single-exchange setting (3.0 shares) when $\gamma = 0$ and $\alpha > 0, \beta > 0$. For a given level of γ , the competition between the market makers is fiercest when $\alpha = 1/2$, which therefore yields the highest equilibrium liquidity supply.

After the trade, we observe the new equilibrium liquidity supply (denoted by the three thicker lines in the bottom of Figure 2). Comparing the liquidity before and after the trade, we observe that a higher fraction of fast traders reduces the buy on ask impact (Proposition 3).

3 Empirical analysis

This section first presents a brief overview of the FTSE 100 stocks' trading environment and data description. Then it explains the DepthAsk(X) and DepthBid(X) liquidity measures and the analysis of the depth impact of trades. It follows with a proxy for the fraction of fast traders, and the empirical analysis of the testable implications of the model.

3.1 Background and data

3.1.1 Market structure of the FTSE 100 stocks

The FTSE 100 stocks are primarily listed on the LSE. In November 2009 the LSE executed 61% of the trading volume (excluding dark pool and over-the-counter volume). The LSE operates an electronic limit order market, with continuous trading between 08:00 and 16:30.

Once stocks are listed on the LSE, alternative trading venues may decide to organize trading in them as well.¹² Four important entrants have emerged that also employ publicly displayed limit order books: Chi-X, Bats, Turquoise, and Nasdaq OMX Europe. While the entrants have the same market model as the LSE, they differ with respect to trading technology (particularly speed), fixed and variable trading fees, and some of the types of orders that may be placed. Investors can demand (“take”) liquidity by placing a market order or supply (“make”) liquidity by placing limit orders at any moment in time. All markets allow for visible, partially hidden (“iceberg”), and fully hidden limit orders.

Chi-X started trading in April 2007 and is the most successful entrant in terms of market share, with 24% of the trading volume in November 2009. Turquoise and Nasdaq OMX started trading FTSE 100 firms as of September 2008, with Bats following two months later. Their market shares are substantially lower, with 5.5%, 1.8%, and 7.6%, respectively. In May 2010, Nasdaq OMX closed down, since they did not meet their targeted market shares.¹³ The five trading venues use identical tick sizes in the sample period, which depend on the stock price. Compared to those in the United States, tick sizes in the United Kingdom are very small, with for example £0.001 for a price between £5.00 and £10.00 and £0.005 for a price between £10.00 and £50.00. All entrants employ a “maker-taker” pricing schedule, where executed limit orders receive a rebate of 0.18 to 0.20 basis

¹²This feature makes the current study inherently different from the literature on cross-listings, where the firms themselves may choose to list on several exchanges.

¹³See “Nasdaq OMX to close pan-European equity MTF,” April 28, 2010, www.thetradenews.com.

points, while market orders are charged 0.28 to 0.30 basis points of the traded value. These make-take fees are very small compared to the tick size of around five basis points on average, and the fees are small compared to those of US stocks.

The trading venues with publicly displayed limit order books execute approximately 60% of the total volume, while the remaining 40% is executed on dark pools, broker-dealer crossing networks, internalized and OTC.¹⁴

3.1.2 Data

The analysis is based on a size-stratified subsample of ten FTSE 100 stocks (see Table 1 for the list). The sample period consists of the 21 trading days in November 2009 and high-frequency data are obtained from the Thomson Reuters Tick History database. For each stock, the data contain separate limit order books for the five trading venues.

For each transaction, the data contain the price, traded quantity and execution time to the millisecond, while for each limit order placement, modification, or cancellation, the data set reports the time stamp and the ten best prevailing bid and offer prices and their associated quantities.

While the time stamp is by the millisecond, we take snapshots of the limit order books at the end of every 100-millisecond interval, resulting in approximately 300,000 observations per day of each stock. Each snapshot contains the outstanding liquidity and trading volume on the buy and sell sides of each venue. The advantage of taking snapshots is that the data are evenly spaced, such that lagged variables in the regressions become easily interpretable.

Fully hidden and partially hidden iceberg limit orders are not directly observable. However, the executed hidden quantity is constructed from the trade size and the state of the order book directly before and after the trade.

¹⁴Source: Fidessa, a trading technology company (see <http://fragmentation.fidessa.com>).

Table upper panel of 1 presents summary statistics of the sample stocks. By construction, there is a large variation in market cap across stocks: The mean and standard deviation are £26 billion and £38 billion respectively. Accordingly, the trading volume (in shares and pounds) and realized volatility also vary substantially. In contrast, the market shares of the five trading venues are fairly stable between stocks and are representative of the entire FTSE 100 index.

The lower panel of Table 1 presents summary statistics of the average number of limit order book modifications and transactions, measured across trading hours for each stock. While the LSE's market share in terms of volume is the largest by far, the number of transactions per hour lies much closer together (i.e., trade sizes are smaller on the entrants). In fact, the average number of limit order modifications on Chi-X (11,690) exceeds that of the LSE (8,610). The limit order books of the entrants are highly active. The ratio of limit order modifications to trades is 32:1 for the LSE and 49:1 for Chi-X, and it increases as a venue's market share goes down to 121:1 for Nasdaq. This is mostly due to high-frequency traders placing and canceling many limit orders. Chi-X and Bats are the most successful entrants in terms of market share, number of transactions, and limit order book activity.

3.2 The DepthAsk(X) and DepthBid(X) liquidity measures

This subsection explains the DepthAsk(X) and DepthBid(X) liquidity measures. The DepthAsk(X) measure aggregates the monetary value of all shares offered at prices between the midpoint and the midpoint plus X basis points. Similarly, the DepthBid(X) measure sums the monetary value of all shares offered within the midpoint and the midpoint minus X basis points. The midpoint is the average of the best bid and ask prices available in the market and I choose $X = 10$ basis points relative to the midpoint. Depending on the tick size, the Depth(10) aggregates the quoted depth of two to five price levels on each side of the order book. The price constraint X guarantees that only depth at

prices close to the midpoint is aggregated, that is, at good price levels. This is important, since liquidity offered deeper in the order book is less likely to be executed and therefore less relevant to investors.

Define by p_j the price level $j = 1, 2, \dots, J$ on the pricing grid and by M the consolidated midpoint. Then for venue ν ,

$$DepthAsk(X)_\nu = \sum_{j=1}^J P_{j,\nu}^{Ask} Q_{j,\nu}^{Ask} \mathbf{1}(P_{j,\nu}^{Ask} < M(1 + X)), \quad (6a)$$

$$DepthBid(X)_\nu = \sum_{j=1}^J P_{j,\nu}^{Bid} Q_{j,\nu}^{Bid} \mathbf{1}(P_{j,\nu}^{Bid} > M(1 - X)). \quad (6b)$$

The measures are calculated at the end of every 100-millisecond interval and represent liquidity offered on the bid and ask sides, per trading venue. For larger values of X , liquidity deeper in the order book is also incorporated.

The depth measure has two features that make it highly suitable for the empirical approach. First, the measure is calculated per venue for the bid and ask sides, which allows us to analyze the correlations between the demand and supply of liquidity across venues, sides and over time. Second, the measure incorporates limit orders beyond the best price levels, making it robust to small, price-improving limit orders. Such orders are often placed by high-frequency traders, who mostly drive the dynamics in the model.

Table 2 contains summary statistics on the Depth(10) measure for the bid and ask sides, reported in British pounds and calculated per venue. The depth measures are calculated per trading hour for each stock, using data sampled at the 100-millisecond frequency. The statistics are calculated across hours and stocks. First and most strikingly, we observe that the Depth(10) offered on Chi-X is 80% of the LSE, while Chi-X executes less than a third of the LSE volume. The liquidity at Bats is roughly 40% of the LSE Depth(10), while Turquoise and Bats have approximately 18% each. On average, the ask and bid sides are

symmetrical.

The regression analysis works with *changes* in $\text{DepthAsk}(10)$ and $\text{DepthBid}(10)$, i.e., the difference between period t and $t - 1$. As Equation (6) shows, these changes depend on changes in the midpoint and limit order book activity (i.e., the placement, cancellation, modification, and execution of limit orders). The model predicts that changes in the depth measures are due to limit order book activity and therefore I define $\text{Chg_DepthAsk}(X)_t$ as the difference in DepthAsk holding the midpoint constant

$$\text{Chg_DepthAsk}(X)_t = \text{DepthAsk}(X, M_{t-1})_t - \text{DepthAsk}(X, M_{t-1})_{t-1}. \quad (7)$$

It is similarly defined for the $\text{Chg_DepthBid}(X)_t$. The measures show how much depth (in British pounds) is added or removed from the limit order book in a period.

3.3 The depth impact of trades

This subsection tests the basic prediction of the model that a trade on one venue is followed by changes in liquidity on the bid and ask sides of all venues. The following model regresses the change in $\text{DepthAsk}(10)$ or $\text{DepthBid}(10)$ of venue N on the contemporaneous and lagged buy and sell volumes of the individual venues $\nu = \{1 \dots 5\}$. The model contains lags up to ten seconds (100 periods), which should leave the market sufficient time to incorporate the information content of trades. Instead of estimating 100 individual lagged coefficients, we estimate coefficients for five groups of lagged variables and enforce that the coefficients are constant within each group. This is obtained by adding in the regression five variables with average lagged trading volume of 1, 2-4, 5-10, 11-20 and 21-100 periods ago, per venue for buy and sell volumes.¹⁵

¹⁵Experimenting with the data reveals that the current definition of the five lagged groups yields the highest predictive power. Although it may seem more natural to estimate a model with an exponential decay parameter, the advantage of the linear model is that the lagged coefficients can easily be averaged and compared across stocks.

The following model is estimated per venue, for each firm-hour. Denote by Buy_{t-l}^ν and $Sell_{t-l}^\nu$ the total buy and sell volumes in interval t for venue $\nu = \{1, \dots, 5\}$ and period $l = \{0, \dots, 5\}$ for the current observation and five lagged groups. Then, for each venue N , we estimate

$$Chg_DepthAsk(10)_t^N = c + \sum_{\nu=1}^5 \sum_{l=0}^5 \left(\beta_{l,\nu}^{Buy} \times Buy_{t-l}^\nu + \beta_{l,\nu}^{Sell} \times Sell_{t-l}^\nu \right) + \varepsilon_t. \quad (8)$$

This model is also estimated for each venue using $Chg_DepthBid(10)_t^N$ as dependent variable. The terms after the constant represent the buy and sell volumes (in British pounds) for the five venues covering the contemporaneous term and the five lagged groups. The coefficients shows how many pounds close to the midpoint are submitted or canceled after a one-pound buy or sell trade on any venue.

All trading venues allow for partially hidden (iceberg) limit orders, where the hidden volume becomes visible after partial execution of the iceberg order (see Section 3.1.2). Because such trades reduce the hidden depth but not the visible depth, it is most natural to subtract the hidden part of the executed trades from Buy_{it} and $Sell_{it}$.¹⁶

Table 3 reports the results averaged over 1,890 regressions. While each regression contains 36,000 observations, note that the parameters are only identified by observations where trades occur. Each column shows the results for one regression model, and displays the contemporaneous coefficients and the cumulative effects after one second and after ten seconds. The first column shows that the immediate effect of a one-pound LSE purchase on LSE DepthAsk(10) is -0.78 pounds. This implies that while the trade removed one pound, either 22 cents is immediately replenished or a new limit order is first placed that immediately provokes the trade. The latter explanation is consistent with the findings of Hasbrouck and Saar (2009).

¹⁶Without this correction, the coefficient $\beta_{0,\nu}^{Buy}$ reflects mechanical changes in visible depth when iceberg orders are executed, and can no longer be interpreted as submissions or cancellations of limit orders.

Consistent with the general prediction of the model, a one-pound buy trade at Chi-X is immediately followed by cancellations on the LSE ask side of -0.198 pounds. After ten seconds, the effect is -0.46, meaning that almost half of the Chi-X trade size is canceled on the LSE. The coefficients for the other venues are of similar magnitude, which range between -0.197 and -0.41 (all significant at the 1% level). These effects are economically large and imply that trades on the entrants are immediately followed by cancellations of limit orders on the same side of the LSE order book. Note that this effect cannot be explained by investors who simultaneously trade on several venues, since the regression controls for trades on all individual venues.

The LSE responds more strongly to Chi-X and Bats trades than vice versa. In columns (2) and (3), LSE trades reduce depth on Chi-X and Bats by -0.33 and -0.17 after ten seconds (which is significantly different from the values -0.458 and -0.31 above). This result can be explained by the model if the entrants attract fewer slow traders. The one-second impact of an LSE trade on the change in liquidity of entrants is -0.62 (the row sum of columns (2) to (5) of Table 3).

A Chi-X buy trade reduces the Chi-X DepthAsk by -1.06 after one second (column (2)) and the liquidity of competitors by -0.86. Define the “Buy on Ask” impact (*BoA*) as the impact of a buy trade on the DepthAsk(10) consolidated over all venues.¹⁷ The *BoA* impact of Chi-X trades is -1.92 and those of the LSE and Bats are -1.48 and -1.65, respectively. These impacts are lower than -1, as predicted by Proposition 2. According to the model, the *BoA* should be -1 in the single-exchange setting. The impact of sell trades on bid side liquidity (columns (6) to (10)) are almost identical to those for buys on ask side liquidity.

The effect of an LSE sell trade on the LSE DepthAsk(10) is 0.545 after 10 seconds (column (1), bottom panel). The coefficients are similar for trades on the other venues

¹⁷The *BoA* is obtained by using as dependent variable the change in consolidated DepthAsk(10) in regression (8), and it corresponds to the row sum of columns (1) to (5) in Table 3.

(e.g., the coefficients of Chi-X and Bats are 0.535 and 0.529, respectively). This result is a basic prediction of the theory that sales convey negative information about the fundamental value of a stock, such that the market makers submit new limit orders on the ask side.

Define the “Buy on Bid” impact (*BoB*) as the impact of a buy trade on the Depth-Bid(10) consolidated over all venues. The *BoB* impact of an LSE sell is 1.43 after ten seconds, which exceeds one as predicted by Proposition 4. The *BoB* impact of Chi-X and Bats trades are even larger (1.87 and 1.82, respectively) and are of the same magnitude as *BoA*. The *BoB* impacts of Turquoise and Nasdaq OMX are smaller (0.89 and 1.09, respectively).

In the subsequent analysis, we average the *BoA* and *BoB* coefficients of the four entrants weighted by market share. First, this reduces the impact of poorly estimated coefficients when only few trades occur on an entrant in a particular hour. Second, it seems natural to combine the entrants given that they have a similar infrastructure in terms of fast trading systems, various order types, small transaction sizes, and trades fees.

Table 5 reports summary statistics for the four impacts (*BoA* and *BoB* for buys and *SoA* and *SoB* for sells). The *BoA* and *SoB* impacts are -1.43 and -1.44 for LSE trades, respectively, and -1.78 and -1.71 for entrants trades. The *BoB* and *SoA* are slightly smaller, ranging between 1.16 and 1.45. Further, the standard errors of the estimates across firm-hours are quite large (ranging between 0.77 and 1.16), which in the model is explained by the variation in the fraction of fast traders.

3.4 Proxy for the fraction of fast traders

Testing the predictions of the model requires a proxy for the fraction of fast traders, which ideally varies across stocks and over time. The fast traders in the model submit two market orders *simultaneously* to both venues. The crucial element is that the market makers

are unable to revise their limit order schedules in between the two trades. Essentially, the second market order executes against limit orders that have not yet incorporated the information content of the first market order. Empirically, this feature is used to identify market orders that are sent “simultaneously” to several venues.¹⁸

Specifically, I use the definition that the current and previous trade occur simultaneously when the following three conditions are satisfied: (i) The best bid and ask prices and quantities of the venue executing the current trade have not changed since the previous trade, (ii) the current trade occurs within 100 milliseconds of the previous trade, and (iii) both trades are either purchases or sales. The first restriction immediately follows from the model and states that the current trade executes against “stale” limit orders.

The following approach identifies simultaneous trades. Per stock, all trades of the five venues are aggregated in a single file,¹⁹ chronologically sorted and timestamped to the millisecond. At each trade we observe the prevailing best bid and ask prices and associated quantities of the five limit order books. Then, the current and previous trade occur simultaneously if above conditions are satisfied. The simultaneous market orders are combined into a single order, as the model states that one fast investor splits up her order across multiple venues.

The first proxy of the fraction of fast traders, denoted $\hat{\gamma}_1$, is the percentage of market orders that occur simultaneously. It is calculated per trading hour of each stock as the average of the dummy variable $D_{i,t}^{Simul}$, which equals one when the trade occurs simultaneously and zero otherwise. The measure corresponds exactly to the model, which assumes that fast investors always split up their trades across venues. In reality however, a fast trader may not access both venues if the entire order can be executed at the best price

¹⁸Note that in the model, market makers always update their quotes after a trade on a competing venue. This holds empirically as well, since 94% of the trades on entrans are followed by changes in the best quotes on the incumbent exchange within one second.

¹⁹In the data, a single market order can execute against several limit orders, generating multiple observations. These partial executions have the same time stamp (at the millisecond level) and are pooled to obtain a single trade.

level of one venue. In this case, the fraction of simultaneous trades depends on the quoted depth, and $\hat{\gamma}_1$ likely underestimates the true fraction of fast traders.²⁰

The second proxy of the fraction of fast traders addresses this issue. Denote the likelihood of a simultaneous trade by $\gamma P(x > T)$, where $P(x > T)$ is the probability that the order x is larger than the size threshold T . This threshold indicates that a fast trader must access two venues and is explained in more detail below. Assuming the order size is exponentially distributed with mean ϕ , we have $P(x > T) = \exp(-T/\phi)$. Then, for all trades t , per trading hour h of stock i , $\gamma_{i,h}$ can be estimated by the linear regression

$$D_{i,t}^{Simul} = \gamma_{2,i,h} P(x > T)_{i,t} + \epsilon_{i,t}, \quad (9)$$

where $P(x > T)_{i,t} = \exp(-\frac{T_{i,t}}{\phi_{i,h}})$ and $\phi_{i,h}$ is the average order size for stock i in trading hour h . The second proxy for the fraction of fast traders $\hat{\gamma}_2$ is the vector of estimated coefficients.²¹

The threshold T is the quoted depth of the most liquid venue at the time of the trade. This assumes that a fast trader only accesses multiple venues if the order cannot be executed on a single venue, which is optimal if venues charge fixed transaction fees. Thus, $T = \text{Max}(Q_{1,1}, \dots, Q_{5,1})$, where $Q_{\nu,1}$ is the number of shares offered on venue ν at price level 1. We take the quantity on the ask side in case of a buy market order and the quantity on the bid size in case of a sell market order. When only a single venue ν offers shares at the best price level, T changes as a fast trader will access two venues when $x > Q_{\nu,1} + Q_{\nu,2}$. With this definition of T , the coefficient $\hat{\gamma}_2$ controls for the nonlinear impact of the quoted depth available at the time of the trade, and therefore it corresponds exactly to the theoretical fraction of fast traders.

²⁰In fact, due to this bias an increase in the quoted depth causes a reduction in $\hat{\gamma}_1$. This causality is the reverse of Proposition 2, where an increase in γ causes a reduction in quoted depth.

²¹There is no constant in this regression because it has no natural interpretation. Adding a constant however does not affect $\hat{\gamma}_2$ much, which has a correlation of 0.91 across firm-hours with the current measure. Further, all the main results are unaffected when using either measure.

Table 4 shows summary statistics of $\hat{\gamma}_1$, the variable $P(x > T)$, and the results of regression (9). The sample average of $\hat{\gamma}_1$ is 0.2, indicating that 20% of the trades are classified as simultaneous. This implies that the fraction of market orders stemming from fast investors is much larger, since they trade on at least two venues. Further, the sample average of $P(x > T)$ is quite low (0.36), which indicates that only 36% of the trades are so large that multiple venues need to be accessed. This explains why the second proxy for the fraction of fast traders is substantially higher, $\hat{\gamma}_2 = 0.40$ on average, after controlling for the available depth at the time of the trade. Further, there appears to be quite some cross-sectional and time-series variation in $\hat{\gamma}_2$, since the standard deviation based on 1,890 firm-hours is 0.1 (all values lie between 0.10 and 0.69). The R-squared of regression (9) is 0.18 on average, which means that, within a given hour, the model seems to have reasonable explanatory power in predicting simultaneous trades. Across firm-hours, the correlation between $\hat{\gamma}_1$ and $\hat{\gamma}_2$ is 0.61.

The proxies for the fraction of fast traders depend on several assumptions, which need further empirical validation. First, the bottom panel of Table 4 shows, for market orders on each venue, the frequency table of the number of venues accessed simultaneously. Traders often access multiple venues simultaneously, since for example 17.3% and 8.1% of the market orders on the LSE are part of simultaneous orders that access two and three venues, respectively. Across all venues, 38.7% of the market orders are part of executions that access two or more venues simultaneously. Also, 2.6% of all market orders are executions part of orders that access the five venues simultaneously. Second, Figure 3 confirms that the order size follows (approximately) the exponential distribution. It plots the order size distribution for trades of the first stock in the sample (Aviva plc), during the first trading hour of November 2, 2009 (left panel) and the whole day (right panel).²² The solid line is the theoretical exponential distribution, which fits the data remarkably well. The results

²²We cannot plot the distribution of all sample trades because the average order size varies over time and across stocks. The sum of multiple exponentially distributed variables is no longer exponentially distributed.

look similar for other stocks and different points in time.²³ Third, Figure 4 plots the distribution of the time between trades, and shows that the requirement that simultaneous trades occur within 100 milliseconds seems reasonable. In fact, about half of all market orders occur within 100 milliseconds of the previous market order.

Our definition of fast traders is similar to the fraction of traders with Smart Order Routing Technology (SORT) of Foucault and Menkveld (2008). They use violations of price priority across markets to identify the fraction of smart routers. Their approach cannot be easily extended to a setting with five trading venues however. Further, the proxy of the current paper fits better with the theoretical model, where market makers face higher adverse selection risk when trading with investors who access multiple venues simultaneously. Yet, the main results hold when using the proxy by Foucault and Menkveld (see Table 1 in the Web appendix).

Note that absent trader identities, we are not certain whether two simultaneous trades indeed stem from the same fast trader. However, this is not an issue, because the adverse selection risk in the model is driven by the size of the market orders and not by the identity of the trader. That is, market makers face the same risk of being picking off when two simultaneous market orders stem from the same or from different traders.

3.5 Testing the propositions: The effect of fast traders

This subsection tests the three propositions of the model that relate the fraction of fast traders (γ) to the Consolidated Depth(10) and the depth impact measures of buys and sells on the change in DepthAsk(10) and DepthBid(10) (defined as the *BoA*, *BoB*, *SoA* and *SoB*).

²³The Kolmogorov-Smirnov test for equality to the exponential distribution is rejected in most trading hours however, most likely due to the large number of trades.

The following regression uses one observation per firm–hour to test the propositions:

$$DepVar_{i,h} = c_i + \delta_{(h)} + \beta_1 \widehat{\gamma}_{i,h} + \beta_2 X_{ih} + \varepsilon_{ih}, \quad (10)$$

where *DepVar* is one of the measures described above; c_i and $\delta_{(h)}$ are firm and daily dummy variables, respectively; and the vector X_{ih} is a set of control variables that affect liquidity which are commonly added in the literature (e.g., Foucault and Menkveld (2008)). The vector X contains the logarithm of turnover, realized volatility (calculated hourly as the sum of the twelve squared five-minute returns), the logarithm of the average order size and a proxy for algorithmic trading. The proxy for algorithmic trading is that of Hendershott, Jones, and Menkveld (2011), and may capture the cross-market activity of market makers. It is defined as the negative of trading volume in hundreds of pounds divided by the number of electronic messages (placements, executions, and cancellations of limit orders). Table 5 contains summary statistics of the regression variables.

The depth impact of trades consists of estimated coefficients and may have substantial measurement error when only a few trades occur in a particular firm–hour. Therefore, regression (10) is estimated by weighted least squares, using as the weight variable the number of trades in each firm–hour.

Table 6 shows the results. Columns (1) and (2) have the logarithm of the Consolidated Depth(10) as the dependent variable and confirm Proposition 2 that a larger fraction of fast traders reduces consolidated depth. A one standard deviation increase in $\widehat{\gamma}_1$ (0.05) reduces the consolidated liquidity by 14%. This result should be interpreted with care, however, since $\widehat{\gamma}_1$ is the fraction of simultaneous trades which in fact depends on the quoted depth (as discussed in section 3.4). The second proxy $\widehat{\gamma}_2$ is specifically designed to measure the fraction of fast traders while controlling for the quoted depth available at the time of each trade. Column (2) shows that a one standard deviation increase in $\widehat{\gamma}_2$ (0.10) reduces consolidated depth by 6.7%, which is significant at the 99% level. The intuition

in the model is that market makers face higher adverse selection risk when trading with fast traders, because conditional upon execution their trades are larger since they may have already traded on the competing venue. The market makers, in turn, scale back their liquidity supply.

Turning to the coefficients of the other variables, it appears that algorithmic trading has no significant effect on consolidated depth. As expected, turnover and the average order size are positively correlated to liquidity, while volatility has a negative coefficient.

Columns (3)-(6) of Table 6 show the effect of the fraction of fast traders on the impact of buy trades on the DepthAsk(10) (*BoA*). It is positive and significant in all specifications, which confirms Proposition 3. The sample average of the *BoA* impact is -1.43, which becomes -1.32 after a one standard deviation increase in $\hat{\gamma}_1$. This effect is economically sizeable, as it implies that a trade is followed by cancellations of limit orders of 32% of the trade size, instead of 43%. The corresponding effect is 0.07 for a one standard deviation increase in $\hat{\gamma}_2$ (column (4)). The intuition in the model is that a lower fraction of fast traders reduces the adverse selection risk faced by the market makers. In response, they increase their liquidity supply across venues, but cancel a larger part of the additional liquidity after a trade on one venue.

Algorithmic trading increases the magnitude of the cancellations across venues. Since the measure depends on the number of limit order submissions and cancellations, it is likely a proxy for the cross-venue activity of high-frequency market makers in the model.

Columns (5) and (6) of Table 6 show the results for the *BoA* impact of entrant trades. The coefficients on $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are approximately 20% larger than those for the LSE. Note that the sample average of the *BoA* impact of entrant trades is also larger (-1.78). Columns (7) to (10) show the results for sell trades on the bid side depth (*SoB*). All coefficients are of similar magnitude as before and statistically significant, which suggests that the mechanisms in the model are symmetric and robust.

Table 7 shows the results for the impact of buy trades on DepthBid(10) (columns (1) to (4)) and of sell trades on DepthAsk(10) (columns (5) to (8)). The impact of a buy trade on bid liquidity is the opposite of its impact on ask liquidity and therefore the signs of all regression variables are the reverse of those in Table 7. Consistent with Proposition 4, the signs of the fraction of fast traders are all negative and significant. A one standard deviation increase in $\hat{\gamma}_1$ reduces the impact of an LSE sell on DepthAsk(10) across venues by 0.10, which is sizable, given a sample average of 1.16. More importantly, the magnitude of these coefficients are similar to those of Table 7. The implication is that trades have an ambiguous effect on liquidity, as a buy trade is followed by cancellations of limit orders on the ask side of competing venues and submissions of limit orders on the bid sides of all venues. The magnitudes of both the cancellations and submissions decline as the fraction of fast investors increases.

A potential issue is that fast traders are also more informed on average, which would explain the negative impact of fast traders on consolidated liquidity.²⁴ However, a Hasbrouck (1991) analysis reveals that fast trades do not have a significantly higher price impact than slow trades (see Table 2 in the Web appendix).

In the model, the fraction of fast traders is exogenous and causally determines the depth and the depth impact of trades. Empirically however, it is possible that the level of depth prevailing at time of the trade affects the decision of fast traders to place simultaneous trades. The second proxy of the fraction of fast traders is designed to control for this effect. Further, only few microstructure theories have predictions about the depth impact of trades, and therefore it is not obvious which control variables should be added in the regressions. While the main results hold for various model specifications, an omitted variables bias may still exist.

²⁴For example, (Chakravarty, Jain, Upson, and Wood, 2012) show that the price impact of Intermarket Sweep Orders (ISO) is higher than that of regular orders. While ISO orders are similar to our definition of fast trades, the difference is that ISO orders are not necessarily executed simultaneously across venues.

4 Conclusion

A friction arises in fragmented equity markets when not all traders may access all trading venues simultaneously. High-frequency market makers partially restore this inefficiency by increasing their liquidity supply across venues. However, after a trade on one venue they cancel the additional liquidity on the other venues. The magnitude of these cancellations depends on the fraction of fast and slow traders in the market.

The model focuses on two trading venues only, but already predicts substantial cancellations of limit orders after a trade. Therefore, the relevance of the model is only strengthened by the fact that most European stocks trade on more than four trading venues (with publicly displayed limit order books) and some US stocks on as many as thirteen trading venues. A larger number of trading venues increases the magnitude of the cancellations following a trade. A drawback of the model is that it does not endogenize the size of the incoming market order, or the decision of traders to adopt fast trading technology.

An important implication of the model is that the cross venue cancellations are efficient and reflect the information content of trades. Recent proposals to curb excessive limit order activity by introducing cancellation fees will reduce liquidity in this model. Such fees hamper competition between trading venues, and in particular harm those venues with a small market share and high cancel-to-trade ratios.

A Appendix

This section contains the proofs of the propositions in the theoretical model. Some proofs contain long and tedious equations, which are relegated to the Web appendix and created using Mathematica.

A.1 Proof of Proposition (1)

The liquidity is in equilibrium when the marginal limit order earns zero expected profits.

We assume that fast traders, when indifferent to trade on either venue, use a “pro-rata” rule and send a fraction $Q_{A1}/(Q_{A1} + Q_{B1})$ of the order to venue A and the complementary fraction to venue B. If the order size exceeds $(Q_{A1} + Q_{B1})$, the pro-rate rule applies to price level 2. Denote by $\bar{F}_q = 1 - F(q)$ the probability that the incoming order x is larger than q ; and by $E_q(V)$ the $E(V|x > q)$. Then, the expected profit of a limit order on price level 1 of venue A, at any location q in the queue of limit orders, is

$$\Pi_{A,q} = \alpha(1 - \gamma)\bar{F}_q(p_1 - E_q(V)) + \gamma\bar{F}_{q(Q_{A1}+Q_{B1})/Q_{B1}}(p_1 - E_{q(Q_{A1}+Q_{B1})/Q_{B1}}(V)).$$

The term $\bar{F}_{q(Q_{A1}+Q_{B1})/Q_{B1}}$ follows from the order splitting rule by the fast traders, who execute limit order q only if $xQ_{A1}/(Q_{A1} + Q_{B1}) > q$, or $x > q(Q_{A1} + Q_{B1})/Q_{B1}$.

The profitability of the marginal limit order is obtained by substituting q for Q_{A1} , and integrating the equation over the distribution of the incoming order x (using Equations (1) and (2)):

$$\begin{aligned} \Pi_{A,Q_{A1}} &= \int_{Q_{A1}}^{\infty} \alpha(1 - \gamma)(p_1 - E(V) - \lambda x) \frac{1}{\phi} \exp\left(-\frac{x}{\phi}\right) dx + \\ &\quad \int_{Q_{A1}+Q_{B1}}^{\infty} \gamma(p_1 - E(V) - \lambda x) \frac{1}{\phi} \exp\left(-\frac{x}{\phi}\right) dx = 0. \end{aligned}$$

The integrals go to infinity, because the marginal limit order is executed for any trade larger or equal to Q_{A1} (first line) or $Q_{A1} + Q_{B1}$ (second line). Solving the integral equation

yields

$$\begin{aligned}\Pi_{A,Q_{A1}} &= \alpha(1 - \gamma)(p_1 - E(V) - \lambda(\phi + Q_{A1})) \exp(-\frac{Q_{A1}}{\phi}) + \\ &\quad \gamma(p_1 - E(V) - \lambda(\phi + Q_{A1} + Q_{B1})) \exp(-\frac{Q_{A1} + Q_{B1}}{\phi}) = 0.\end{aligned}$$

The zero expected profit condition implies that the first line of the last equation is positive and the second line negative. In equilibrium, the market makers on venue A expect to lose to fast traders who access both venues: conditional upon executing against a fast trader, their trades are larger in expectation. This loss is compensated by the expected profits from slow traders who go to venue A first (fraction $\alpha(1 - \gamma)$). The profit equation is solved similarly for venue B, and the two equations with two unknowns are solved yielding Proposition 2.

Existence: By Brouwer's fixed point theorem, a solution exists (see Theorem M.I.1, p.952, in Mas-Colell, Whinston and Green, Microeconomic Theory, Oxford University Press). Define the set of possible solutions for Q_{A1} and Q_{B1} by $S \subset \mathbb{R}^+ \mathbb{R}^+$. This set is compact, as the value of Q_{A1} and Q_{B1} never exceed Q_1^S (the quantity of the single exchange setting). This set is also convex and non-empty. Furthermore, the solution is continuous and differentiable, and thus fulfills the requirements of the theorem.

Uniqueness: From the model solution (3)-(5), define $fa[x]$ as the response of Q_{A1} for a given realization $Q_{B1}=x$, and $fb[x]$ the response of Q_{B1} for a given realization of $Q_{A1}=x$. We show uniqueness by proof of contradiction. Suppose there are two fixed points, $x1 = \{xa1, xb1\}$ and $x2 = \{xa2, xb2\}$, where $x1 \neq x2$. Thus, by definition of a fixed point, we have $xa1 = fa[xb1]$, $xb1 = fb[xa1]$ and $xa2 = fa[xb2]$, $xb2 = fb[xa2]$. It implies that a combination of the parameters exists (γ, α and ϕ , on their relevant domain), for which the

Euclidean distance:

$$|\{xa1, xb1\}, \{xa2, xb2\}| = |\{fa[xb1], fb[xa1]\}, \{fa[xb2], fb[xa2]\}| \quad (A1)$$

$$\sqrt{(xa1 - xa2)^2 + (xb1 - xb2)^2} = \sqrt{(fb[xa1] - fb[xa2])^2 + (fa[xb1] - fa[xb2])^2}$$

The above equations may only hold if the following two equations are both satisfied:

$$(xa1 - xa2)^2 = (fb[xa1] - fb[xa2])^2 \equiv (g[xa1] - g[xa2])^2 \quad (A2)$$

$$(xb1 - xb2)^2 = (fa[xb1] - fa[xb2])^2 \equiv (h[xa1] - h[xa2])^2 \quad (A3)$$

$$\text{where } g[x] = \frac{\gamma}{\gamma + \alpha(1 - \gamma) \exp(\frac{x}{\phi})} x \quad \text{and} \quad h[x] = \frac{\gamma}{\gamma + (1 - \alpha)(1 - \gamma) \exp(\frac{x}{\phi})} x.$$

The function $g[x]$ is similar to $fb[x]$ except that the term Q_1^S drops out. Because $g[x] < x, \forall \{x > 0, 0 < \alpha < 1, 0 < \gamma < 1\}$, we show below that $(xa1 - xa2)^2 \geq (g[xa1] - g[xa2])^2$, and therefore Equation (A2) holds if and only if $xa1 = xa2$. This violates the assumptions that the coordinates of the two fixed points are different, and proofs that the solution is unique. The argument holds symmetrically for (A3), and therefore (A1) does not hold.

To see that $(xa1 - xa2)^2 \geq (g[xa1] - g[xa2])^2$, denote $g[xa1] - g[xa2] \equiv c1xa1 - c2xa2$, where $c1 = \gamma / (\exp(xa1/\phi)(\alpha - 1)(\gamma - 1) + \gamma)$ and $c2 = \gamma / (\exp(xa2/\phi)(\alpha - 1)(\gamma - 1) + \gamma)$. Without loss of generality, assume $xa1 > xa2$, such that $c1 > c2$ and $0 < c1 < 1, 0 < c2 < 1$. Then, $g[xa1] - g[xa2] \leq c2xa1 - c2xa2 < xa1 - xa2$, where the last inequality holds because $c2 < 1$. This proof is also provided in the Web appendix, where all functions are evaluated.

A.2 Proof of Proposition (2)

The proposition contains three claims. First, when $\alpha = 0$ or $\beta = 0$ one venue attracts all liquidity and obtains the depth of the single-exchange setting; otherwise both venues have a strictly positive expected market share and “coexist.”

Proof. Formally, the unique solution of Equations (4) and (5) is $Q_{A1} = 0, Q_{B1} = Q_1^S$ if

$\alpha = 0, \gamma > 0$; and it is $Q_{A1} = Q_1^S, Q_{B1} = 0$ if $\alpha = 1, \gamma > 0$.

Intuitively, suppose $\alpha = 0$ and $\gamma > 0$. Only fast traders have access to venue A, whereas both fast and slow traders have access to venue B. The fast traders execute the marginal limit order on venue A if $x > Q_{A1} + Q_{B1}$. This implies that the marginal limit order on A will only be executed after all shares on venue B are bought. In equilibrium the marginal limit order on venue B earns zero expected profits, and therefore any limit order on venue A must have a negative expected profit. A negative expected profit is not an equilibrium, and therefore $Q_{A1} = 0$ and venue B obtains the liquidity of the single-exchange setting. By symmetry, the argument holds when $\beta = 1 - \alpha = 0$ and $\gamma > 0$.

If $\alpha > 0$ and $\beta > 0$, the market makers will always place a positive amount on both markets. This follows from Equations (4) and (5), where an equilibrium with $Q_{A1} = 0$, $\alpha > 0$ and $\gamma > 0$ does not exist when $Q_1^S > 0$. We analyze the best ask price level which is the lowest price level that attracts a positive number of shares, and thus exclude the trivial case where $Q_1^S = 0$. ■

The second claim states that an increasing fraction of fast traders (γ) strictly reduces consolidated depth.

Proof. We need to show that $\partial(Q_{A1} + Q_{B1})/\partial\gamma < 0$. As we have an implicit solution, we calculate the derivative using the implicit function theorem and Cramer's rule and evaluate its sign. Denote the consolidated depth by $Q_{Cons1} = Q_{A1} + Q_{B1}$, and partial derivatives as e.g. $\frac{\partial g1}{\partial Q_{A1}} = g1_{Q_{A1}}$. Consider the system of implicit equations that are constructed to equal zero:

$$\begin{aligned} g1 &= Q_1^S - \frac{\gamma}{\gamma + \alpha(1 - \gamma) \exp(\frac{Q_{B1}}{\phi})} Q_{B1} - Q_{A1} \\ g2 &= Q_1^S - \frac{\gamma}{\gamma + (1 - \alpha)(1 - \gamma) \exp(\frac{Q_{A1}}{\phi})} Q_{A1} - Q_{B1} \\ g3 &= Q_{A1} + Q_{B1} - Q_{Cons1} \end{aligned}$$

Using the implicit function theorem and Cramer's rule, we have

$$\frac{\partial(Q_{Cons1})}{\partial\gamma} = - \frac{\det \begin{pmatrix} g1_{Q_{A1}} & g1_{Q_{B1}} & g1_{\gamma} \\ g2_{Q_{A1}} & g2_{Q_{B1}} & g2_{\gamma} \\ g3_{Q_{A1}} & g3_{Q_{B1}} & g3_{\gamma} \end{pmatrix}}{\det \begin{pmatrix} g1_{Q_{A1}} & g1_{Q_{B1}} & g1_{Q_{Cons1}} \\ g2_{Q_{A1}} & g2_{Q_{B1}} & g2_{Q_{Cons1}} \\ g3_{Q_{A1}} & g3_{Q_{B1}} & g3_{Q_{Cons1}} \end{pmatrix}}. \quad (A4)$$

The resulting equation is very long and tedious, but it is negative as shown in the Web appendix using Mathematica.²⁵ ■

Third, the consolidated liquidity in a fragmented market is strictly larger than the liquidity in the single-exchange setting.

Proof. We repeat Equations (4) and (5) and rewrite as

$$Q_{A1} = Q_1^S - \frac{\gamma}{\gamma + \alpha(1 - \gamma) \exp(\frac{Q_{B1}}{\phi})} Q_{B1} \equiv Q_1^S - c_{B1} Q_{B1} \quad (A3a)$$

$$Q_{B1} = Q_1^S - \frac{\gamma}{\gamma + (1 - \alpha)(1 - \gamma) \exp(\frac{Q_{A1}}{\phi})} Q_{A1} \equiv Q_1^S - c_{A1} Q_{A1}. \quad (A3b)$$

We need to show that $Q_{A1} + Q_{B1} > Q_1^S$, which is equivalent to $c_{B1} Q_{B1} + c_{A1} Q_{A1} < Q_1^S$.

$$\begin{aligned} c_{B1} Q_{B1} + c_{A1} Q_{A1} &= c_{B1} Q_{B1} + c_{A1} (Q_1^S - c_{B1} Q_{B1}) \\ &= c_{A1} Q_1^S + (1 - c_{A1}) c_{B1} Q_{B1} \\ &< c_{A1} Q_1^S + (1 - c_{A1}) Q_1^S = Q_1^S. \end{aligned}$$

The first equality simply replaces Q_{A1} with the first line of Equation (A3a); which is then rewritten in the second line. The inequality holds because $c_{B1} Q_{B1} < Q_1^S$, since $c_{B1} < 1$ and $Q_{B1} \leq Q_1^S$. ■

²⁵The Web appendix is available at <https://www.dropbox.com/s/upoip0p5wl3ziri/WA.pdf>.

A.3 Proof of Proposition (3)

The proposition contains a definition and three claims. The first claim that the Buy on Ask (BoA) impact of a trade equals -1 in the single-exchange setting is explained in footnote 9. The second claim states that when both venues coexist, $BoA < -1$.

Proof. We must show that $BoA = \lambda \frac{\partial(Q_{A1} + Q_{B1})}{\partial E(V)} < -1$. First, note that a trade affects the equilibrium depth only via its impact on the market makers' expected valuation $E(V)$. It is a linear function of the order size with price impact parameter λ per traded unit (Equation (2)). For a one-unit trade, the effect is $BoA = \partial(Q_{A1} + Q_{B1})/\partial E(V) \times \lambda$. We set up the system in Equation (A4) to calculate $\frac{\partial(Q_{Cons1})}{\partial E(V)} \times \lambda$. The resulting equation is tedious, but the Web appendix shows it is strictly negative when $Q_{A1} < \phi, Q_{B1} < \phi$. ■

Note that the BoA impact is the marginal effect of a trade on the change in quoted depth. For a trade x large enough such that $E(V_t) + \lambda x > p_1$, the trade depletes all quoted liquidity and no additional shares will be cancelled on price level 1. In this case, the BoA is biased towards minus one. Section 3 of the Web appendix shows that all proofs also hold when the depth of price level 1 and 2 are aggregated. This is important for the empirical analysis, since the probability that a trade consumes liquidity of 2 price levels is smaller.

The third claim states that the BoA impact strictly increases by the fraction of fast traders (γ): $\frac{\partial BoA}{\partial \gamma} > 0$, for the parameter range $Q_{A1} < \phi, Q_{B1} < \phi$. The Web appendix confirms that the derivative is negative. A Mathematica file used for numerical simulations confirms that all the hypotheses hold.²⁶

The proofs of Proposition (4) are symmetrical to Proposition (3) and therefore omitted. The reason is that the impact of a trade on liquidity is only driven through its impact on the expected valuation of the market makers.

²⁶All files can be found here: <https://www.dropbox.com/s/9ikur6gynsmxa3n/Simulations.zip?dl=0>.

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Table (1) Summary statistics of the sample firms.

This table presents summary statistics for the 10 FTSE 100 sample stocks using November 2009 data. The upper panel shows statistics across the 10 stocks for market cap (in billions), the price (British pounds), the hourly trading volume (in thousands of shares), hourly turnover (in millions of British pounds), realized volatility (the hourly sum of five minute squared returns), and the market shares of the five trading venues. The trading venues are the LSE, Chi-X, Bats Trading, Turquoise and Nasdaq OMX Europe. The lower panel shows order book data, where statistics are equally weighted based on one observation per trading hour for each stock (1,890 observations). It shows statistics on the number of transactions and limit order book modifications. The stocks in our sample are Aviva, Hsbc Holdings Plc, Itv Plc, Kingfisher, Lonmin, National Grid, Pearson, Sage group, Vedanta Resources and Xstrata.

	Mean	Median	Stdev	Min	Max	Count
Stock characteristics						
Market cap	26.3	10.0	38.3	1.2	129.0	10
Price	6.87	5.29	6.50	0.50	23.24	10
Volume (shares)	12.3	9.7	12.3	2.3	42.8	10
Turnover (British pounds)	7.2	3.9	9.7	0.9	30.6	10
Realized Volatility	1.70	2.67	4.28	0.00	13.81	10
Share LSE	55.59	56.20	5.05	48.41	62.87	10
Share Chi-X	28.80	27.88	3.33	25.28	34.54	10
Share Bats	7.15	6.78	1.62	4.95	10.37	10
Share Turquoise	5.71	5.81	1.66	2.86	8.86	10
Share Nasdaq	2.74	2.90	1.14	1.14	4.18	10
Order book data						
Trades LSE	269	159	329	0	3,698	1,890
Trades Chi-X	241	150	271	0	2,336	1,890
Trades Bats	100	61	121	0	1,458	1,890
Trades Turquoise	64	41	75	0	960	1,890
Trades Nasdaq	31	19	38	0	502	1,890
Limits LSE	8,610	5,722	8,374	0	80,812	1,890
Limits Chi-X	11,690	6,656	14,484	81	149,743	1,890
Limits Bats	6,100	2,964	8,604	19	88,539	1,890
Limits Turquoise	5,312	3,562	5,558	60	55,721	1,890
Limits Nasdaq	3,770	2,073	5,367	0	75,088	1,890

Table (2) Summary statistics of the DepthAsk(X) and DepthBid(X) measures.

The DepthAsk(X) measure aggregates the monetary value of all shares offered at prices between the midpoint and the midpoint plus X basis points. Similarly, DepthBid(X) sums the monetary value of all shares offered between the midpoint and the midpoint minus X basis points. The depth measures are calculated per trading hour with data sampled at the 100-millisecond frequency. The statistics below are calculated equally weighted across hours and stocks (1,890 observations).

<i>Ask side</i>	Mean	Median	Stdev	Min	Max	Count
Depth(10) LSE	78,258	48,969	106,498	0	2,126,085	1,890
Depth(10) Chi-X	65,828	38,288	76,438	0	564,117	1,890
Depth(10) Bats	30,535	16,525	37,421	0	292,518	1,890
Depth(10) Turquoise	14,721	11,463	13,577	0	149,987	1,890
Depth(10) Nasdaq	11,814	5,818	17,974	0	121,374	1,890
<i>Bid side</i>						
Depth(10) LSE	80,031	48,675	206,943	0	7,971,022	1,890
Depth(10) Chi-X	64,942	37,472	75,319	0	577,423	1,890
Depth(10) Bats	30,234	15,816	37,061	0	237,978	1,890
Depth(10) Turquoise	15,104	11,152	17,251	0	392,841	1,890
Depth(10) Nasdaq	12,725	6,342	19,094	0	161,044	1,890

Table (3) The cumulative impact of a trade on Depth(10).

This table shows the cumulative effect over time of the buy and sell turnover of the individual venues on changes in the DepthAsk(10) or DepthBid(10) of one venue. The effects are estimated by regression (8) per hour for each stock (1,890 times), with data sampled at the 100-millisecond frequency. The table reports the average coefficients and displays the contemporaneous effect, and the cumulative effect after one second and ten seconds. Changes in DepthAsk(10) and DepthBid(10) reflect changes in depth offered at prices in the interval [Midpoint, Midpoint $\pm$ 10bps]. These changes stem from limit order book activity (placements, cancellations and executions of limit orders). The independent variables are the contemporaneous and lagged buy and sell trading volumes of the five venues, denominated in British pounds. Hidden executions are subtracted from the trading volume. Standard errors are clustered per firm-hour (not reported) and a single asterisk indicates significance at the 1% level.

<i>£ Buys</i>	Sec	(1)	(2)	(3) DepthAsk(10)		(4)	(5)	(6)	(7)	(8) DepthBid(10)		(9)	(10)
		LSE	Chi-X	Bats	Turq.	Nasdaq		LSE	Chi-X	Bats	Turq.	Nasdaq	
LSE	0	-0.783*	-0.235*	-0.088*	-0.029*	-0.021*		0.290*	0.269*	0.109*	0.040*	0.018*	
LSE	1	-0.792*	-0.309*	-0.166*	-0.068*	-0.072*		0.401*	0.369*	0.196*	0.098*	0.067*	
LSE	10	-0.803*	-0.332*	-0.165*	-0.088*	-0.090*		0.514*	0.436*	0.218*	0.121*	0.085*	
Chi-X	0	-0.198*	-1.034*	-0.134*	-0.015*	-0.022*		0.209*	0.488*	0.141*	0.033*	0.014*	
Chi-X	1	-0.389*	-1.060*	-0.302*	-0.062*	-0.108*		0.367*	0.661*	0.304*	0.131*	0.089*	
Chi-X	10	-0.458*	-1.004*	-0.289*	-0.086*	-0.124*		0.518*	0.761*	0.340*	0.165*	0.112*	
Bats	0	-0.197*	-0.331*	-0.985*	-0.076*	-0.063*		0.269*	0.397*	0.300*	0.067*	0.047*	
Bats	1	-0.258*	-0.368*	-0.860*	-0.093*	-0.124*		0.386*	0.509*	0.381*	0.118*	0.110*	
Bats	10	-0.309*	-0.358*	-0.812*	-0.117*	-0.135*		0.540*	0.686*	0.459*	0.167*	0.153*	
Turq	0	-0.098*	-0.074*	-0.087*	-0.663*	-0.032*		0.062*	0.082*	0.099*	0.078*	0.027*	
Turq	1	-0.150*	-0.088*	-0.098*	-0.659*	-0.039*		0.153*	0.209*	0.142*	0.103*	0.050*	
Turq	10	-0.197*	-0.164*	-0.141*	-0.697*	-0.050*		0.185*	0.286*	0.191*	0.095*	0.064*	
Nasdaq	0	-0.183*	-0.032	-0.125*	-0.110*	-0.853*		0.235*	0.212*	0.243*	0.141*	0.103*	
Nasdaq	1	-0.284*	-0.137	-0.169*	-0.136*	-0.820*		0.264*	0.280*	0.260*	0.161*	0.153*	
Nasdaq	10	-0.409*	-0.091	-0.070	-0.121	-0.849*		0.452*	0.348*	0.301*	0.215*	0.169*	
<i>£ Sells</i>													
LSE	0	0.295*	0.267*	0.106*	0.040*	0.019*		-0.771*	-0.227*	-0.085*	-0.025*	-0.023*	
LSE	1	0.416*	0.375*	0.198*	0.103*	0.070*		-0.795*	-0.311*	-0.169*	-0.065*	-0.074*	
LSE	10	0.545*	0.440*	0.224*	0.126*	0.084*		-0.806*	-0.323*	-0.165*	-0.086*	-0.097*	
Chi-X	0	0.210*	0.460*	0.128*	0.030*	0.014*		-0.189*	-1.014*	-0.130*	-0.017*	-0.020*	
Chi-X	1	0.377*	0.629*	0.282*	0.123*	0.079*		-0.380*	-1.035*	-0.287*	-0.055*	-0.100*	
Chi-X	10	0.535*	0.749*	0.327*	0.151*	0.107*		-0.449*	-0.989*	-0.278*	-0.087*	-0.123*	
Bats	0	0.251*	0.383*	0.264*	0.073*	0.042*		-0.155*	-0.322*	-1.009*	-0.050*	-0.078*	
Bats	1	0.364*	0.508*	0.334*	0.125*	0.061*		-0.222*	-0.349*	-0.878*	-0.067*	-0.112*	
Bats	10	0.529*	0.637*	0.408*	0.152*	0.090*		-0.263*	-0.373*	-0.814*	-0.099*	-0.125*	
Turq	0	0.110*	0.124*	0.123*	0.062*	0.037*		-0.108*	-0.062*	-0.068*	-0.651*	-0.033*	
Turq	1	0.174*	0.204*	0.134*	0.104*	0.061*		-0.098*	-0.035	-0.054*	-0.657*	-0.026*	
Turq	10	0.282*	0.271*	0.160*	0.112*	0.062*		-0.177*	-0.070	-0.052	-0.689*	-0.046*	
Nasdaq	0	0.222*	0.166*	0.199*	0.112*	0.089*		-0.154*	-0.080	-0.120*	-0.101*	-0.830*	
Nasdaq	1	0.247*	0.204*	0.199*	0.133*	0.110*		-0.307*	-0.154*	-0.126*	-0.161*	-0.830*	
Nasdaq	10	0.427*	0.193	0.235	0.169	0.067		-0.448*	-0.114	-0.092	-0.129	-0.845*	
R-squared		0.174	0.178	0.148	0.083	0.096		0.171	0.167	0.133	0.071	0.088	

Each column contains coefficients averaged over 1,890 regressions, where each regression has 36,000 observations.

Table (4) Descriptive statistics of the fraction of fast traders.

The first proxy of the fraction of fast traders ($\hat{\gamma}_1$) is the hourly average of $D_{i,t}^{Simul}$, a dummy variable that equals one if two trades occur *simultaneously* and zero otherwise. The current and previous trades occur simultaneously when (i) the best bid and ask prices and the quantities of the venue executing the current trade have not changed since the previous trade, (ii) the current and previous trades occur within 100 milliseconds of each other, and (iii) both trades are either purchases or sales. Market orders that occur simultaneously are combined into a single order. The regression below is estimated per order and adjusts for the impact of the quoted depth on the classification of simultaneous trades. Denote the probability that the incoming trade is larger than the threshold T by $P(x > T_{i,t}) = \exp(-T_{i,t}/\phi_{i,h})$, where $T = \text{Max}(Q_{1,1}, \dots, Q_{5,1})$. Then T is the maximum of the quoted depth on the five venues, and indicates the order size threshold where a fast trader must access more than one trading venue to prevent a trade-through. We use the depth on the bid side in case of a sell market order and the depth on the ask side in case of a buy market order. The variable $\phi_{i,h}$ is the average order size in trading hour h of stock i . The second proxy for the fraction of fast traders is the estimated coefficient of regression

$$D_{i,t}^{Simul} = \gamma_{2,i,h} P(x > T)_{i,t} + \epsilon_{i,t}. \quad (9)$$

The regression is estimated per trading hour for each stock (1,890 times). The vector $\hat{\gamma}_2$ contains the coefficients $\gamma_{2,i,h}$, and is winsorized at the 1% and 99% levels. The bottom panel shows, for market orders on each venue, the frequency table of the number of venues accessed simultaneously. Columns (1) to (5) show fractions of market orders, and column (6) the total number of market orders on each venue.

Panel A:	Mean	Median	Stdev	Min	Max	Count
$\hat{\gamma}_1$	0.20	0.20	0.05	0.00	0.50	1,890
$P(x > T)$	0.36	0.38	0.09	0.00	0.56	1,890
Regression (9):						
$\hat{\gamma}_2$	0.40	0.40	0.10	0.10	0.69	1,890
R-Squared	0.18	0.20	0.03	0.13	0.21	1,890
Panel B:	Venues accessed simultaneously					
Fraction of trades	1	2	3	4	5	Total
LSE	69.8	17.3	8.1	3.4	1.5	251,381
Chi-X	61.0	23.1	10.1	4.1	1.6	215,876
Bats	48.9	26.4	14.7	7.0	3.0	90,200
Turquoise	53.6	24.2	12.6	6.7	2.9	64,224
Nasdaq OMX	47.4	25.5	14.5	7.9	4.7	30,103

Table (5) Descriptive statistics of the cumulative Depth(10) impact of trades.

The top and middle panels show the cumulative impact over one second of a buy or sell trade on the change in DepthAsk(10) or DepthBid(10) for all venues. These impacts are estimated per hour for each stock by regression (8) for trades on each of the five venues. The table reports the results for trades on the LSE and the average of the four entrants (volume weighted). The estimated variables are winsorized at the 1% and 99% levels. The bottom panel shows trading characteristics measured across hours for each stock. Algorithmic trading is defined as the negative of trading volume (in hundreds of pounds) divided by the number of electronic messages. Order size represents the average transaction size (in £10,000s), and Turnover is the total hourly trading volume (in millions of pounds), summed over all venues. Realized volatility is defined as the sum of squared five minute returns, measured at an hourly frequency.

	Mean	Median	Stdev	Min	Max	Count
Impact on Depth(10)						
Buy on Ask, LSE	-1.43	-1.33	0.77	-3.93	0.00	1,890
Sell on Bid, LSE	-1.44	-1.37	0.82	-3.95	0.04	1,890
Buy on Ask, Entrants	-1.78	-1.65	1.01	-9.60	3.77	1,890
Sell on Bid, Entrants	-1.71	-1.65	0.94	-6.82	1.79	1,890
Buy on Bid, LSE	1.16	0.96	0.88	-0.19	4.23	1,890
Sell on Ask, LSE	1.19	1.01	0.90	-0.23	4.34	1,890
Buy on Bid, Entrants	1.45	1.28	1.16	-3.36	8.42	1,890
Sell on Ask, Entrants	1.36	1.22	1.08	-5.08	6.08	1,890
Hourly trading characteristics						
Algo trading	-1.76	-1.38	1.49	-25.23	-0.03	,1890
Order size	0.97	0.90	0.44	0.12	2.34	1,890
Turnover	7.23	2.86	12.40	0.00	169.00	1,890
Realized volatility	0.02	0.00	0.12	0.00	0.97	1,890

This table shows the impact of the fraction of fast traders (γ) on the Consolidated Depth(10) and the Buy on Ask impact (*BoA*) and Sell on Bid (*SoB*) impact of trades. The Consolidated Depth(10) (columns (1) and (2)) is the sum of the DepthAsk(10) and DepthBid(10) across all venues, and the regression uses the logarithm of the hourly average over 100-millisecond intervals. The variable *BoA* (columns (3) to (7)) is the impact accumulated over one second of one venue’s buy trades on the DepthAsk(10) of all venues and the *SoB* (columns (7) to (10)) is the impact of sell trades on DepthBid(10) of all venues. The impacts are estimated for trades on each of the five venues by equation (8), but we report the results for trades on the LSE and the average of the four entrants (volume weighted). The regressions are based on 1,890 firm–hours and use as weighting variable the number of transactions per firm–hour. The regressions use two proxies for the fraction of fast traders ($\hat{\gamma}_1$ and $\hat{\gamma}_2$), which are described in Table 4. The other independent variables are a proxy for algorithmic trading, the logarithm of the average order size (Ln Order Size), the logarithm of total turnover (Ln Turnover), and the logarithm of volatility based on five minute squared returns (Ln Volatility). See Table 5 for the summary statistics on all variables. The regressions contain firm and day fixed effects. The t-statistics are shown below the coefficients, and are based on robust Newey–West standard errors (heteroscedasticity and autocorrelation robust (HAC)) with nine lags.

[illegible]

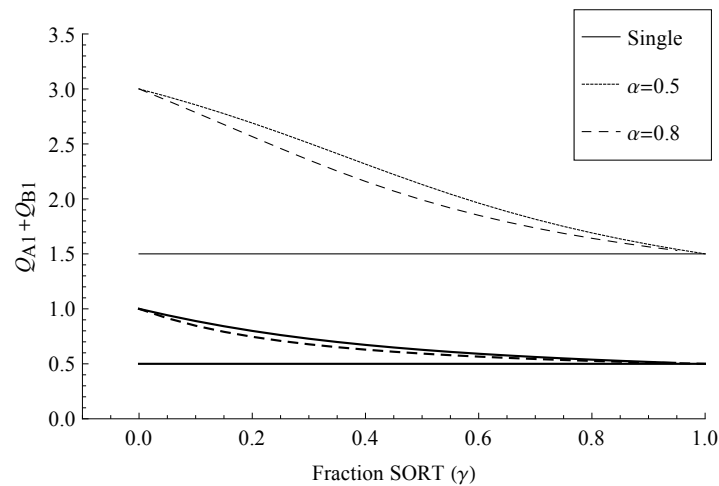
Table (7) The impact of the fraction of fast traders on the Depth(10) impact of trades.

This table shows the impact of the fraction of fast traders (γ) on Consolidated Depth(10) and the Buy on Bid impact (*BoB*) and Sell on Ask impact (*SoA*) of trades. The *BoB* (columns (1) to (4)) is the impact accumulated over one second of one venue’s buy trades on the DepthBid(10) of all venues and the *SoA* (columns (5) to (8)) is the impact of sell trades on the DepthAsk(10) of all venues. The impacts are estimated for trades on each of the five venues by equation (8), but we report results for trades on the LSE and the average of the four entrants (volume weighted). The regressions are based on 1,890 firm–hours and use as weighting variable the number of transactions per firm–hour. The regressions use two proxies for the fraction of fast traders ($\hat{\gamma}_1$ and $\hat{\gamma}_2$), which are described in Table 4. The other independent variables are a proxy for algorithmic trading, the logarithm of the average order size (Ln Order Size), the logarithm of total turnover (Ln Turnover), and the logarithm of volatility based on five minute squared returns (Ln Volatility). See Table 5 for the summary statistics of all the variables. The regressions contain firm and day fixed effects. The t-statistics are shown below the coefficients and are based on robust Newey-West standard errors (HAC) with nine lags.

[illegible]

Figure (2) Numerical Example: Consolidated depth before and after a one-share buy trade.

This figure plots consolidated depth on the best ask price of venues A and B ($Q_{A1} + Q_{B1}$) against the fraction of fast traders γ , for $\alpha = 0.5$ (dotted line), $\alpha = 0.8$ (dashed line), and the base case (solid line). The three thinner lines at the top represent the depth before the one-unit buy trade, and the three thicker lines at the bottom represent the depth after the trade. The fraction of slow traders is $(1 - \gamma)$, of which a fraction α has only access to venue A and $\beta = 1 - \alpha$ has only access to venue B. The single-exchange liquidity obtains when $\alpha = 0$ or $\beta = 0$. The remaining model parameters are held fixed. The average trade size is one, with a per unit price impact of 20 basis points. The fundamental value is £10.00 and the best ask price is £10.005.



Distribution order size for Aviva

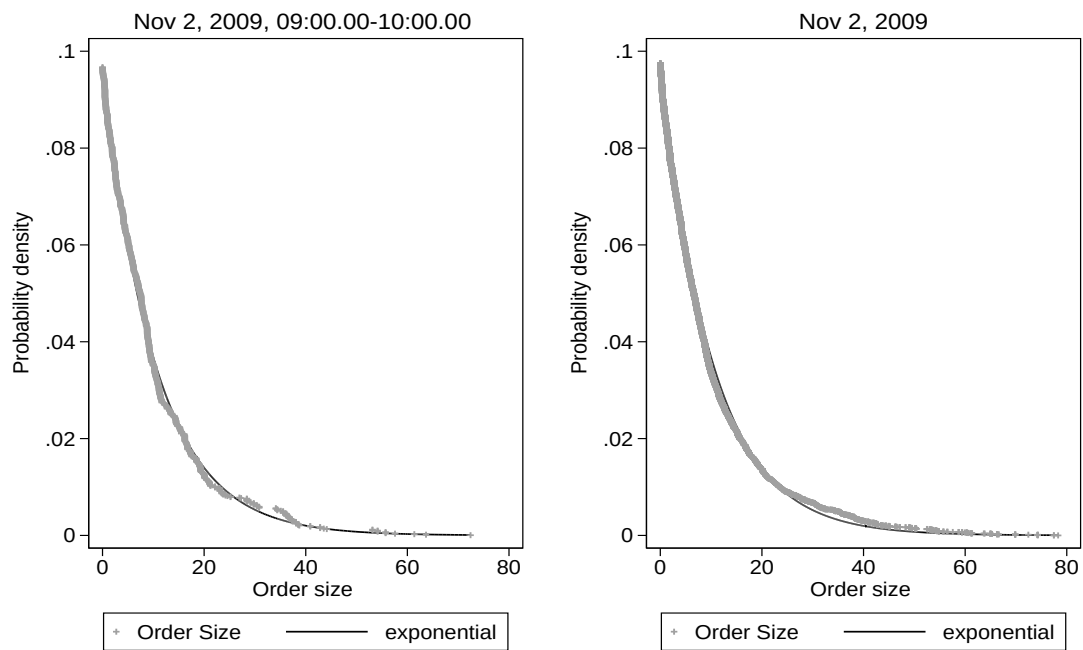


Figure (3) Empirical distribution of the order size.

The figure plots the distribution of the order size (in thousands of British pounds), for trades in the stock Aviva. The grey marks are the empirical realizations and the black line is the fitted theoretical exponential distribution. The left figure shows the distribution for trades in the first trading hour of November 2, 2009, and the right figure shows the distribution for the whole day. Note that when two market orders occur simultaneously their sizes are combined, as they are assumed to stem from one fast trader.

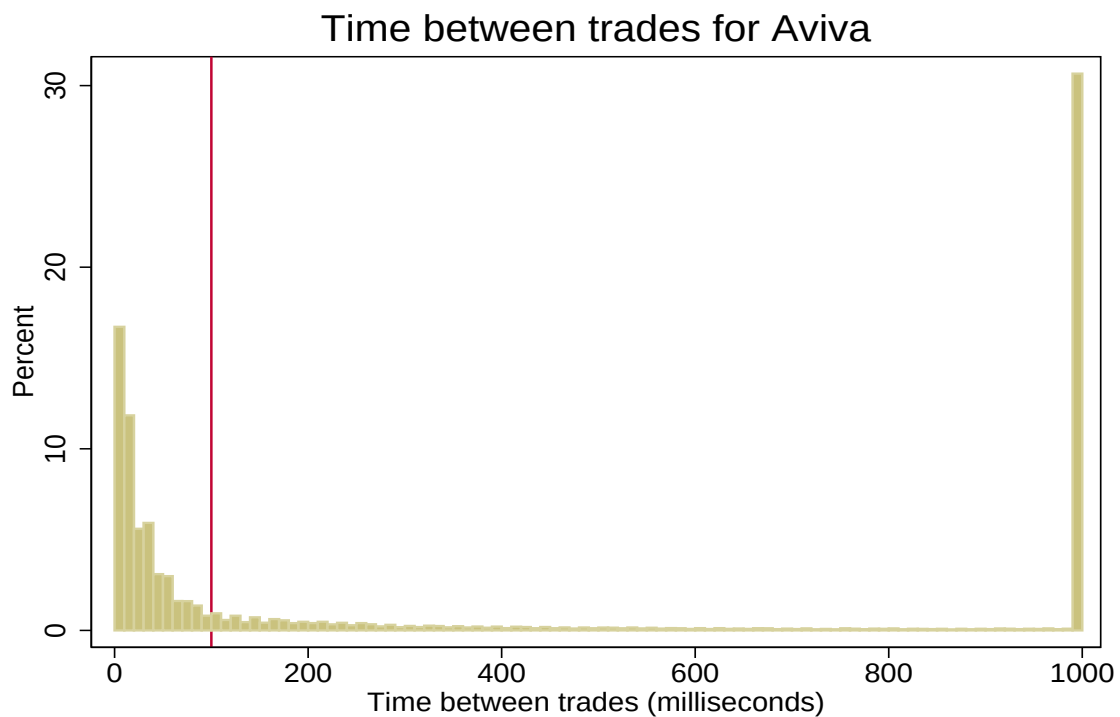


Figure (4) Distribution of the time between trades.

The figure shows the histogram of time (in milliseconds) between market orders on all venues in November 2009 of the stock Aviva. Each bar represents a 10 milliseconds interval, and the vertical red line is placed at 100 milliseconds. To focus on high-frequency activity, the time difference is winsorized at one second (the last bar indicates that 33% of the trades have a time difference exceeding one second).